1.2 Exploring RStudio

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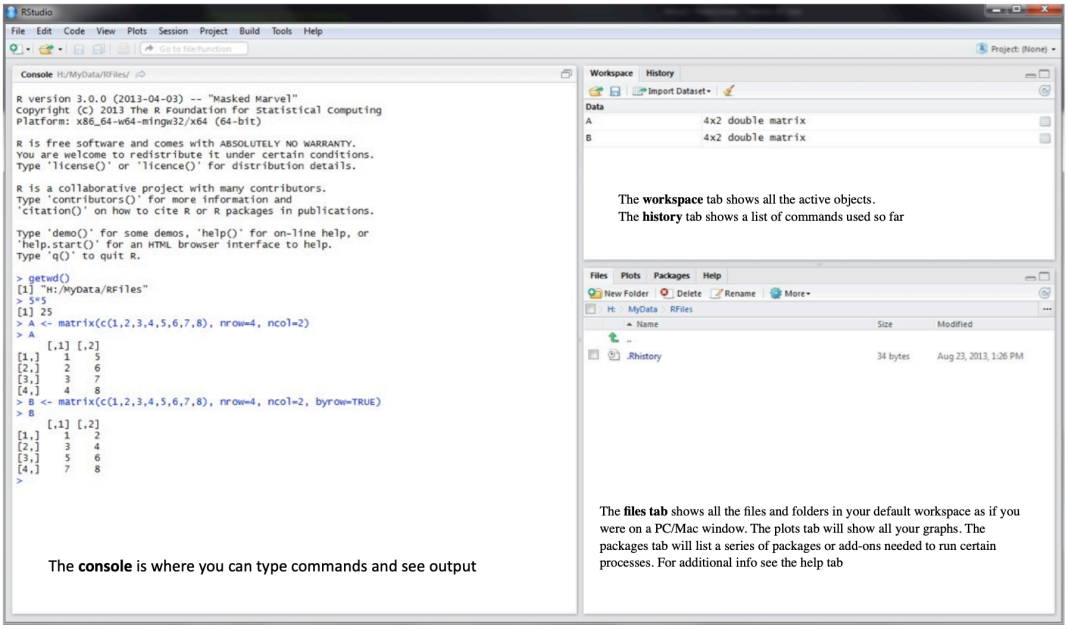
Introduction to the RStudio interface, illustrating how to interact with R via the console, store and reuse values using variables, and preserve your work using R scripts.

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The RStudio interface

Once you've opened RStudio, you will see several empty panes similar to this:



Console

Let's focus on the **console** window on the left. This is where we can directly communicate with R by entering "code". The way this code is written follows certain arbitrary conventions – just like natural languages such as English or German. Here is an example in the R programming language:

```
print("Hello world!")
```

```
[1] "Hello world!"
```

Entering this command into your console and pressing ENTER will display the sentence "Hello world!" in a new line.

As we can see, anything that R should understand as a simple sequence of letters or words must be enclosed by quotation marks `"..."`. Anything inside them will be interpreted as a so-called **string**. Their counterpart are **numbers** or **integers**, as illustrated here:

```
2 + 2 - 1
```

```
[1] 3
```

Naturally, you can use R for more sophisticated computations:

- $2 + (3 \times 3)$

```
2 + (3 * 3)
```

```
[1] 11
```

- $\sqrt{9}$

```
sqrt(9)
```

```
[1] 3
```

- $\frac{16}{2^3}$

```
16 / 2 ^ 3
```

```
[1] 2
```

Working environment

While it is certainly useful to print text or numbers to your console, it may sometimes make more sense to (at least temporarily) store them somewhere, so you can re-use them later. In fact, R gives us a way of storing data in virtual, container-like objects: **variables**. We can assign strings or numbers to a variable by using the assignment operator `<-`.

When you run the commands below, you will (hopefully) notice two items popping up in your **Environment/Workspace** tab in the top right corner.

```
greeting <- "Hello world!"

quick_math <- 2 + 2 - 1
```

If we now want to display the content in the console, we can simply apply the `print()` function to the variable:

```
print(greeting)
```

```
[1] "Hello world!"
```

```
print(quick_math)
```

```
[1] 3
```

We can also embed variables in other statements. For example, let's take the content of `quick_math` and multiply it with 2.

```
hard_math <- quick_math * 2

print(hard_math)
```

```
[1] 6
```

R Scripts

Working with the console has a very plain, yet important disadvantage: Once we close RStudio, the console is wiped clean, erasing everything we've typed into it during our precious R session.

The remedy for this nuisance are **scripts**. Essentially, a script is the programmer's equivalent of a Word document: It allows you to save all the code you've written to a file, which you can seamlessly continue working on the next time you open it.

To **create a new R script** you can either go to:

- "File" → "New" → "R Script" or …
- click on the icon with the + sign and select "R Script" …
- or simply press Ctrl+Shift+N (MacOS: Cmd+Shift+N)

Don't forget to save your script with Ctrl+S (Cmd+S)! It is good practice to save your files regularly.